Grade 11 Physics: Kinematics and Motion in One Dimension

1. Introduction to Motion

Motion is defined as a change in position of an object over time. To describe motion accurately, we use physical

Key Concepts & Definitions:

- Position (x): Location of an object relative to origin.
- Displacement (Delta x): Shortest distance from initial position (xi) to final position (xf).
  $\Delta x = x_f - x_i$
- Velocity (v): Rate of change of displacement with respect to time.
  $v_{avg} = \Delta x / \Delta t$
- Acceleration (a): Rate of change of velocity with respect to time.
  $a_{avg} = \Delta v / \Delta t$

2. Equations of Motion for Uniform Acceleration

1) $v = u + at$
2) $s = ut + 0.5 * a * t^2$
3) $v^2 = u^2 + 2 * a * s$

Where:

u = Initial velocity (m/s)
v = Final velocity (m/s)
a = Constant acceleration (m/s^2)
t = Time duration (seconds)
s = Total displacement (meters)

3. Solved Example Problem

Problem: A ball is dropped from a height of 45 meters. Calculate time taken to reach the ground and its final velo

Solution:

$s = ut + 0.5 * a * t^2$

$45 = 0 + 5 * t^2 \Rightarrow t^2 = 9 \Rightarrow t = 3$ seconds.

$v = u + at = 0 + 10 * 3 = 30$ m/s.